

Riley Weber

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Interdisciplinary full-stack developer and graphic designer who blends creative design expertise with strong technical foundations. Adept communicator who collaborates across disciplines to deliver visually compelling, scalable, robustly tested web solutions that convey complex and technical information in an accessible format.

SUMMARY OF SKILLS

- **Frontend Development:** 3 years' experience with **TypeScript, Angular, JavaScript (ES6+), HTML5, CSS3, Sass, Node.js, NPM**, and responsive web design.
- **Systems Development:** 8 years' experience building native applications using **C, C++, and Python**.
- **Graphic & UI Design:** Strong portfolio in **UI/UX design**, using HTML, CSS, SASS, and Bootstrap to create visually engaging and user-friendly interfaces that communicate highly technical information at a glance.
- **DevOps and Infrastructure:** Docker, GitLab CI, Github Actions, PostgreSQL, MySQL, MongoDB, REST APIs, SpringBoot.
- **Performance & Systems:** Profiling (LIKWID), low-latency design, optimization with Intel Intrinsics, OpenMP, and OpenMPI.
- **Other Tools & Platforms:** CUDA, Go, Java, Bash, Linux (Debian, Ubuntu, RHEL, Arch, NixOS), TensorFlow, scikit-learn, SWIG, clang++, make, cmake, RabbitMQ, MQTT.

EXPERIENCE

Sep 2021 – Sep 2023

Engineering Scientist Associate, University of Texas at Austin – Applied Research Laboratories

- Doubled the performance of a mission-critical C++ message passing library. Added compatibility for approximately 10 new types in the SWIG Python interface for this product, responding directly to the needs of Python developers.
- Improved software written in C++ and Python that provides real-time GPS data to the department of defense.
- Managed the creation of GitLab continuous integration pipelines from the ground up. Used Docker to automate building, testing, code health report generation, and packaging for two high-priority applications.

Jan 2020 – Jun 2021

Research Assistant, University of North Carolina at Charlotte

- Crafted a visualization combining utilitarian design and attractive components to communicate complex performance data.
- Managed an individual C/C++ project to create a custom profiling software using hardware counters (model-specific registers) which analyzed architecture performance on sparse graph problems.
- Used Intel intrinsics, OpenMP, and likwid-perfctr to benchmark desktop systems and compute clusters.

May 2019 – Aug 2019

IT Intern, Sealed Air – Innovation & Development

- Spearheaded the design and creation of a userflow designed to inform users of our rebranded URL. Developed the full stack, from REST APIs written in Java to customer-facing UX written in JavaScript.
- Automated data collection in Python and R for a visualization used to communicate common fault codes to senior executives. This enabled visual analysis of thousands of data points as recent as the current minute.
- Used JavaScript, Angular, CSS, HTML, R, Python, RabbitMQ, ThingWorx, and Java.

Jan 2017 – Dec 2017

Computer Support Representative, Brigham Young University – Center for Teaching and Learning

- Managed conversion of paper processes to TypeScript Angular web applications improving efficiency for two departments and over 110 employees. Used both Apache/php/MySQL (LAMP) and MongoDB/Express/Angular/NodeJS (MEAN) stacks.
- Maintained 50 Debian/Ubuntu servers which provided access to over 200 websites for many departments across campus.
- Sourced and supported hardware for animators, graphic designers, video editors, and programmers.

EDUCATION

M.S. in Computer Science, Systems Emphasis, University of North Carolina at Charlotte – Jun 2021

B.S. in Computer Science, Systems Emphasis, Minor in Mathematics, University of North Carolina at Charlotte – Dec 2020